Chapter 1
Introduction to IoT

What is IoT?

- Internet of Things is the network of smart physical objects:
  - physical objects (e.g. devices, vehicles, buildings, etc.) embedded with sensors, computation unit, memory unit, power source, and network connectivity,
  - which enables the physical object to collect and exchange data,
  - analyze the collected data to extract new insight and respond accordingly.
- Goal of IoT is to “connect the unconnected”
  - “Things” or “objects” that were not supposed to be connected to the Internet
  - IoT did the technology transition in traditional computer networks

What is IoT?

- IoT is the Unifications of technologies:
  - Embedded Systems
  - Low Power and Low Rate Network
  - Internet
  - Cloud Computing
  - Data Analytics
  - Bigdata
  - Edge Intelligence
  - Network Security and Data Security
  - Software Defined Networks

What is IoT?

- Alternate Definition:
  - “The Internet of Things (IoT) is the network of physical objects that contain embedded technology to communicate and sense or interact with their internal states or the external environment.”

Brief History of Initial Phase of IoT

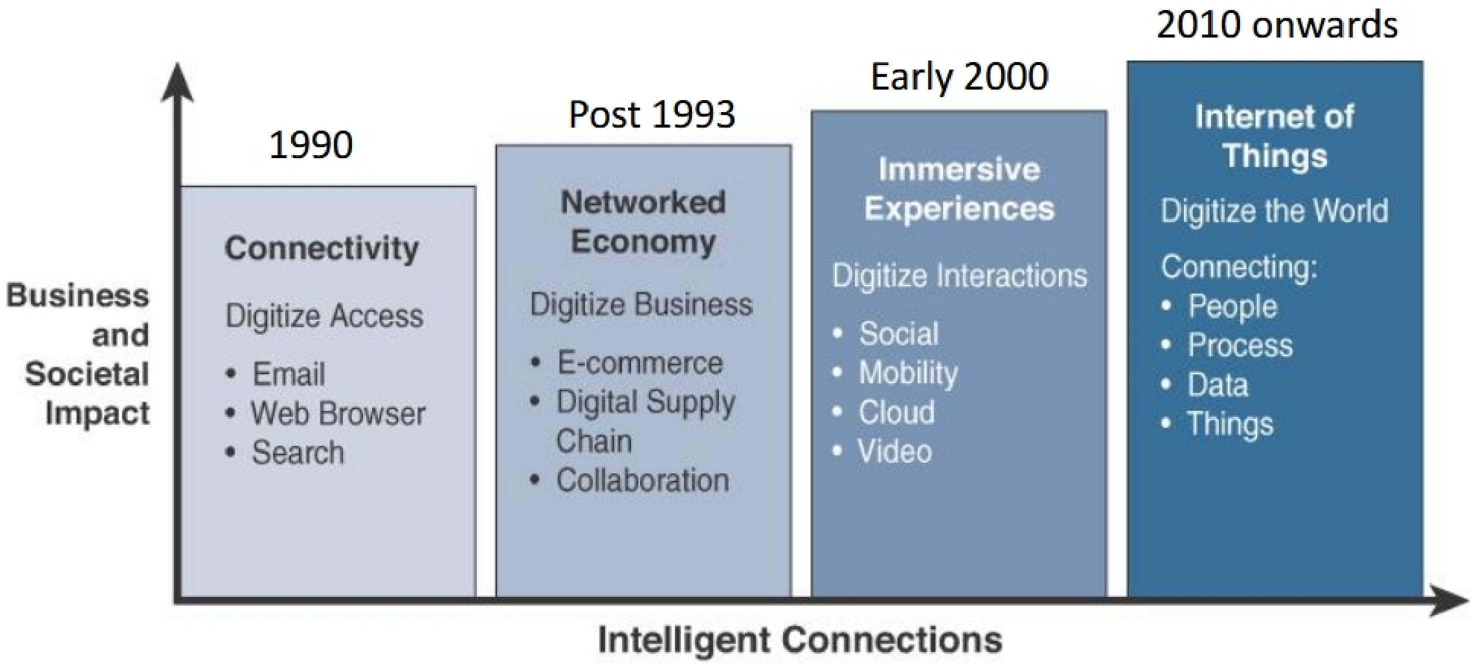
- The term "Internet of Things" was likely coined by Kevin Ashton of Procter & Gamble, later MIT's Auto-ID Center, in 1999.
- "In 20th century, computers were brains without senses — they only knew what we told them."
- "Now in 21st century, computers are sensing things for themselves!"

Brief History of Initial Phase of IoT

- Early 1980s at the Carnegie Melon University, a group of students created a way to get their campus Coca-Cola vending machine to report on its contents through a network in order to save them the trek if the machine was out of Coke.
- In 1990, John Romkey, developer of the first TCP/IP stack for IBM PC in 1983, connected a toaster to the internet for the first time.
- In 1991, a group of students at the University of Cambridge used a web camera to report on coffee available in their computer labs coffee pot.
- At the beginning of the 21st Century, LG Electronics introduced the world’s first refrigerator connected to the internet

Brief History of Initial Phase of IoT

- The popularity of IoT did not accelerate until 2010/2011 and reached mass market from 2013-14.
- Definition of the IoT has also evolved over time.



Benefits of IoT

- Real-Time Monitoring
  - IoT sensors can monitor equipment and processes in real-time, providing valuable sensed data that can be used to streamline operations, reduce waste, and increase output.
- Automation of Processes
  - Machines can assemble parts with more precision and speed, resulting in fewer errors during assembly
  - Robots can very rapidly detect faults that may not be detected by the human eye

Benefits of IoT

- Improved Efficiency or Productivity
  - IoT helps to enhance productivity by streamlining processes, automating tasks, and providing real-time data insights..
- Predictive Maintenance
  - Machines can assemble parts with more precision and speed, resulting in fewer errors during assembly
  - Continuous monitoring of systems and processes to identify key indicators of problems before they result in downtime or system failure

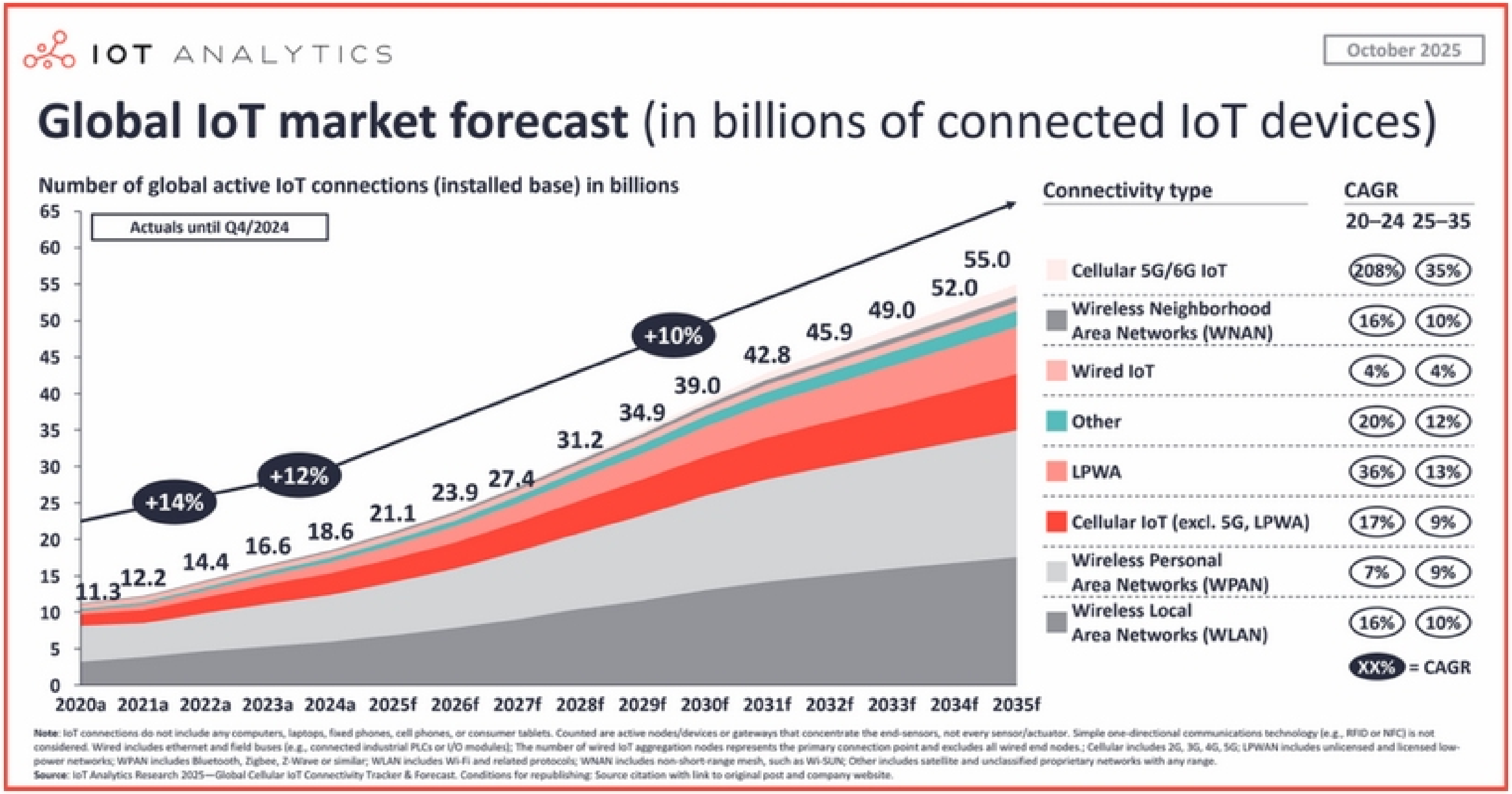
Benefits of IoT

- Improved or New Insights
  - IoT generates vast amounts of data that can be analyzed to identify trends, inefficiencies, and areas for improvement. Organizations can leverage these insights to enhance productivity and operational effectiveness.
- Cost Reduction
  - When an organization can improve system uptime, automate processes, reduce the risk of failure and gain insights that support better decision making, and reduce resource usage, the result is efficiency and cost savings.

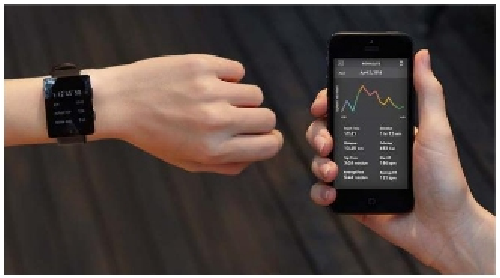
Benefits of IoT

- Optimized Work Environments
  - IoT technology can help create more comfortable and efficient workspaces
- Adaptability
  - The ability to adapt to new business requirements, customer needs, and changing conditions, or scale the deployment in response to business growth or customer requirements

Growth of IoT Devices



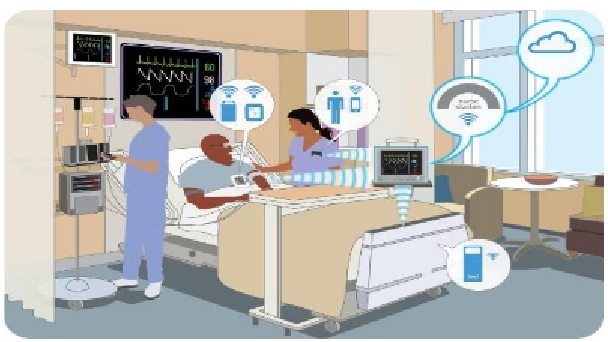
Where is IoT?



Wearable
Tech Devices

Smart Appliances

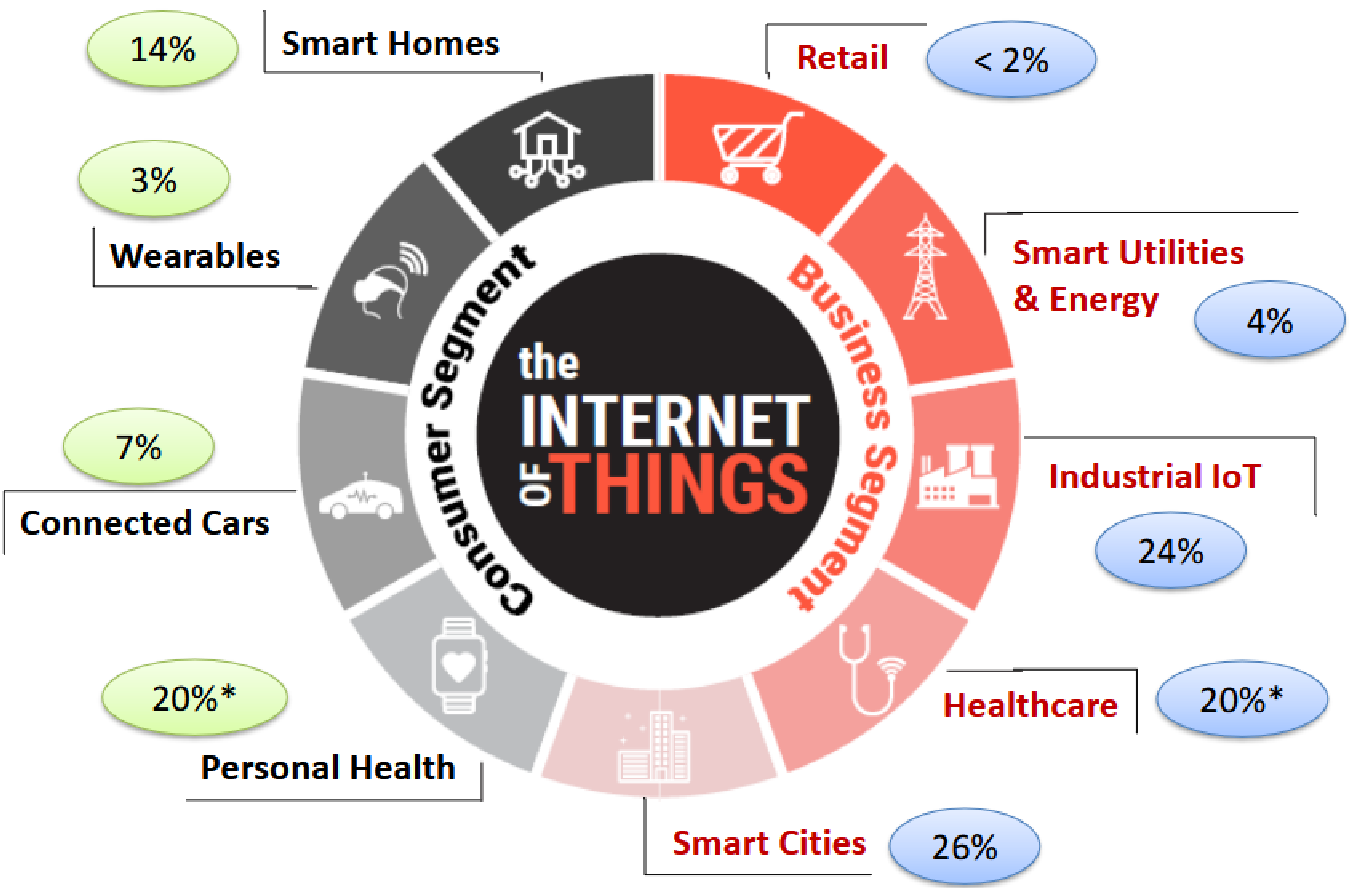
It's everywhere!



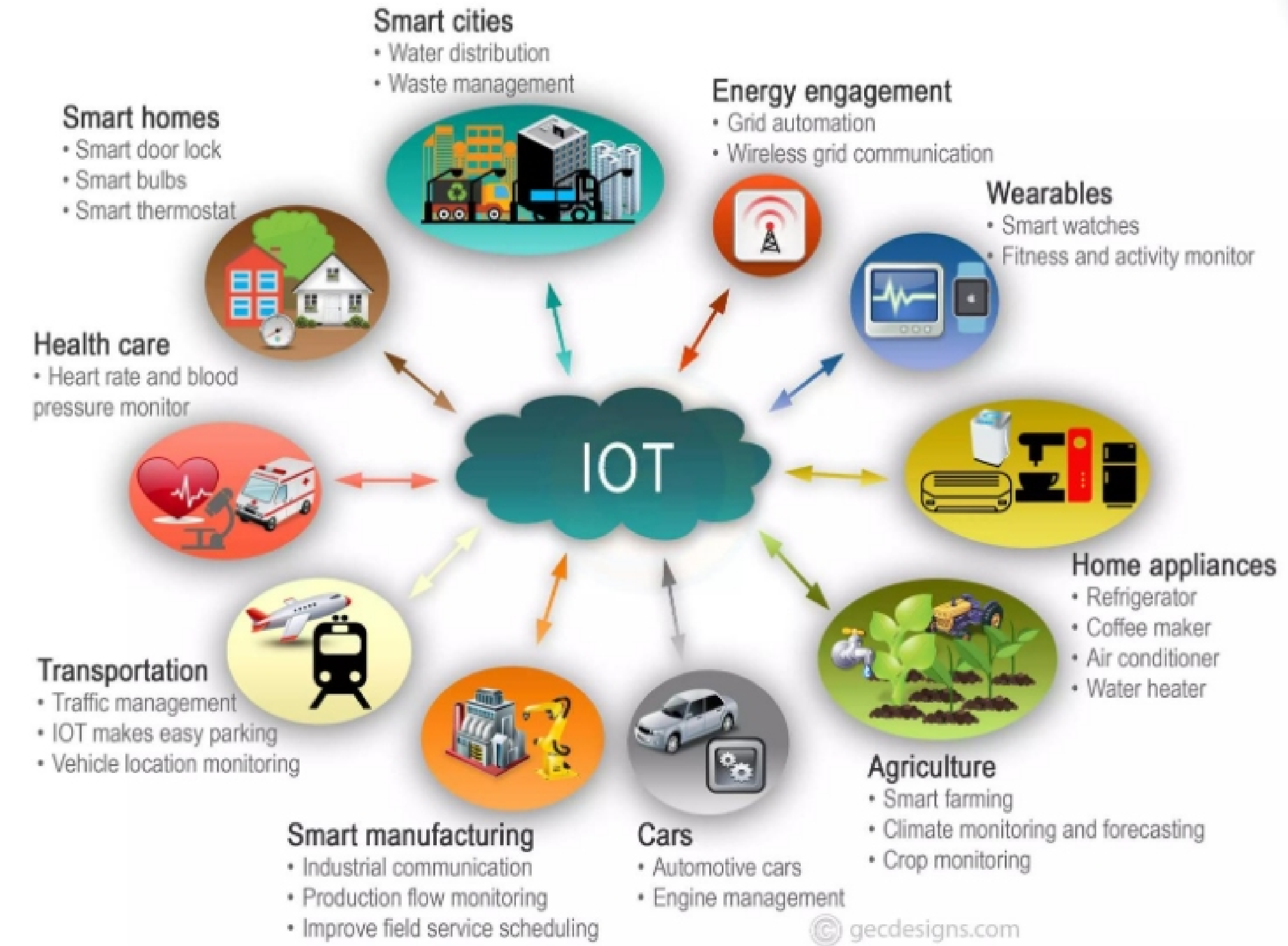
Healthcare

Industry Automation
and Monitoring

Global IoT Market Share



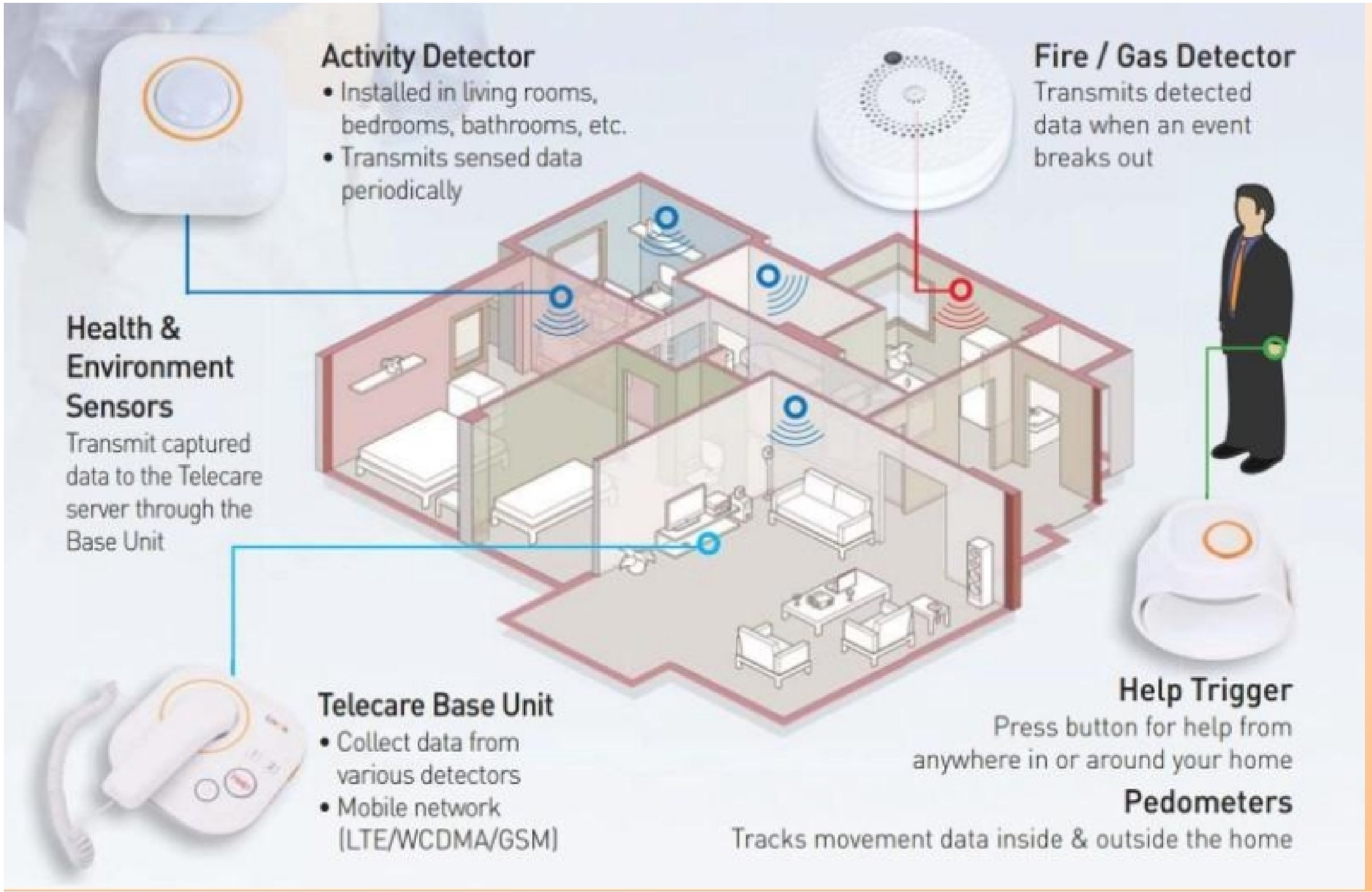
Applications of IoT



Smart City

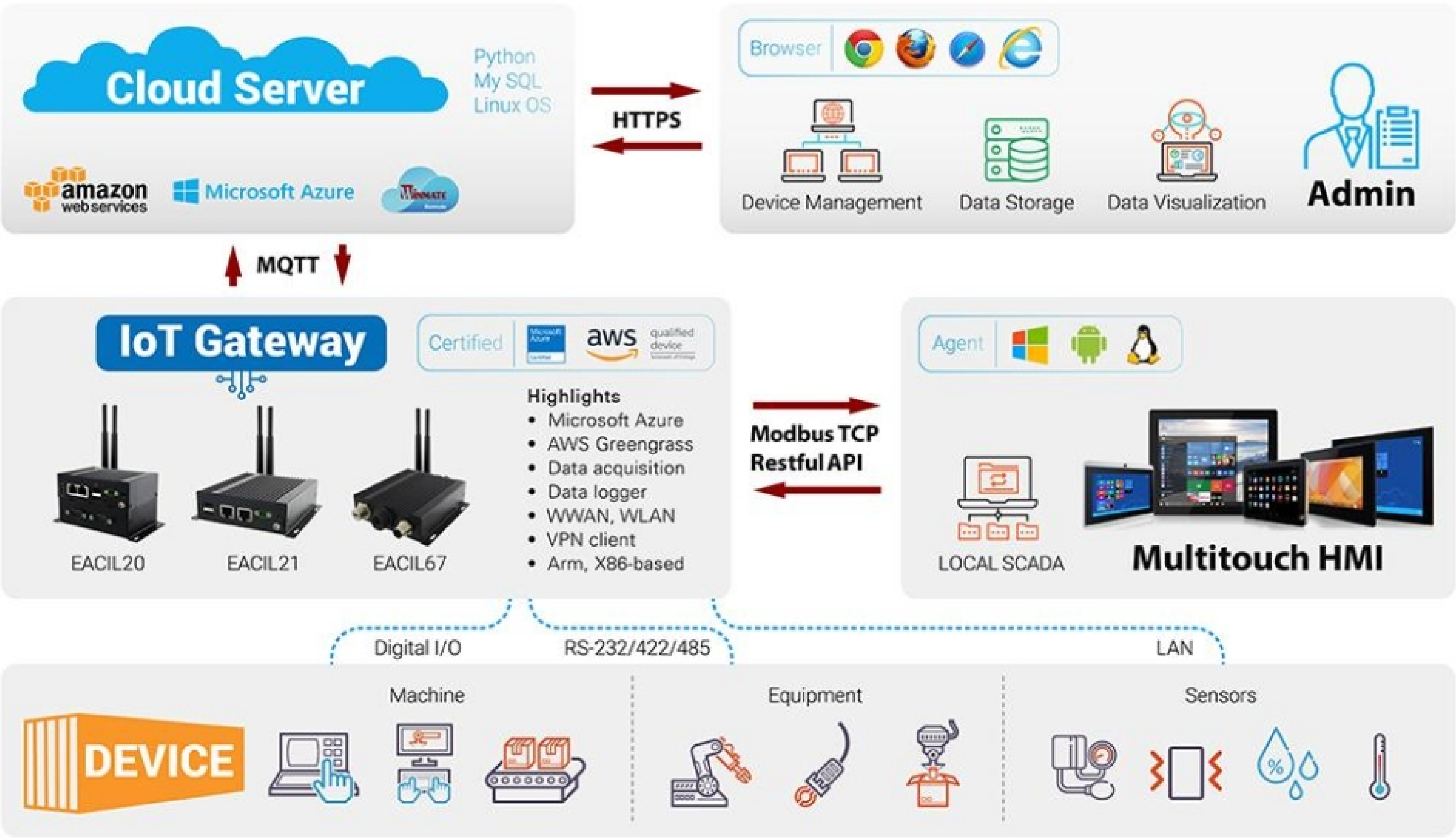
Traffic Management
Education
Air Pollution
Open Data
Internet of Things
Smart Health
Intelligent Shopping
Smart Environment
Electromagnetic Emissions
Smart Buildings
Public Safety
Smart Home
Gas & Water Leak Detection
Smart Street Lights
Smart Energy
Electric Vehicle Charging
Water Quality
Smart Parking
Waste Management

Smart Home

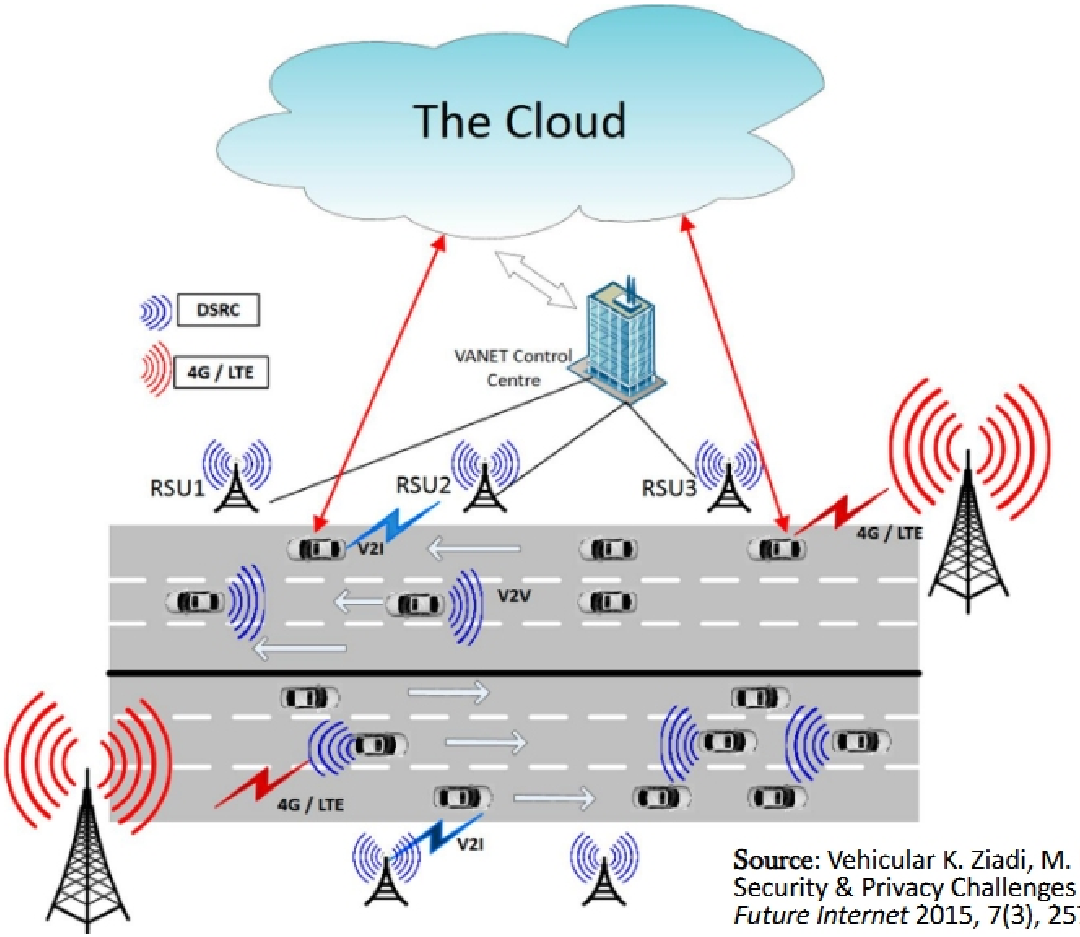


Activity Detector
• Installed in living rooms, bedrooms, bathrooms, etc.
• Transmits sensed data periodically
Fire / Gas Detector
Transmits detected data when an event breaks out
Health & Environment Sensors
Transmit captured data to the Telecare server through the Base Unit
Telecare Base Unit
• Collect data from various detectors
• Mobile network (LTE/WCDMA/GSM)
Help Trigger
Press button for help from anywhere in or around your home
Pedometers
Tracks movement data inside & outside the home

Industrial IoT

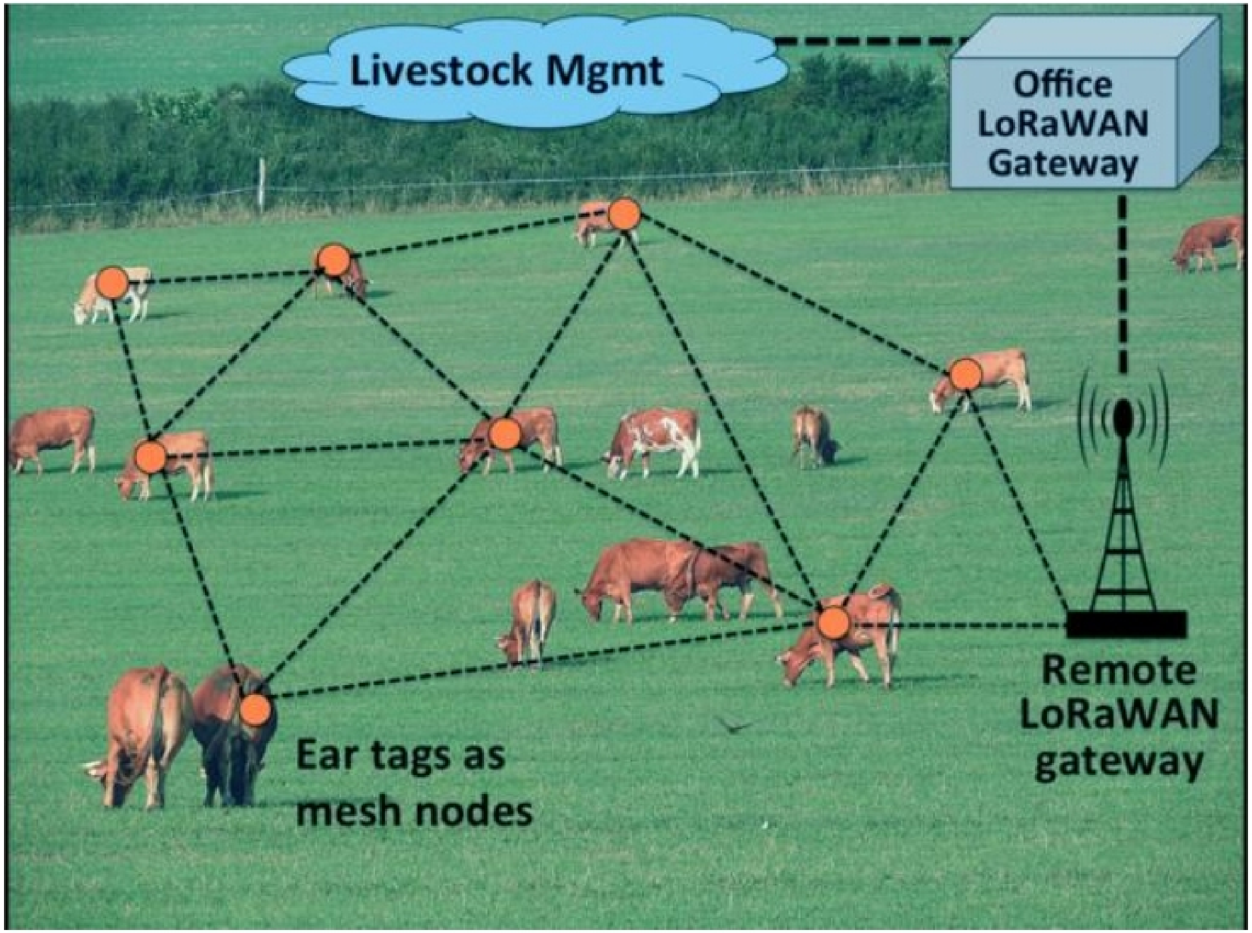


Connected Cars



Source: Vehicular K. Ziadi, M. Rajarajan, "Internet: Security & Privacy Challenges and Opportunities", *Future Internet* 2015, 7(3), 257-275.

Livestock Management



Main Challenges in IoT Implementation

Sensors

- limited resources
- limited types of sensors

Scale

- millions of devices are connected to form IoT

Low Power Network

- devices should remain connected to the network for years
- high network latency
- can't use traditional communication protocols

Interoperability

- various protocol, various architecture
- unavailability of standardized platform
- different technology leads to interoperability issue

Bigdata & Data analytics

- massive amount of sensor data
- different sources and various forms
- extract intelligence form the heaps of data

Privacy

- which personal data to share with whom
- how to control

Security

- "things" becomes connected, so security becomes complex

IoT Ecosystem and Architecture

- It is an interconnected network of physical devices, communication infrastructure, computing resources (both edge and cloud), applications, data, and stakeholders that collaboratively collect, exchange, and process data to enable intelligent service.

IoT Ecosystem and Architecture

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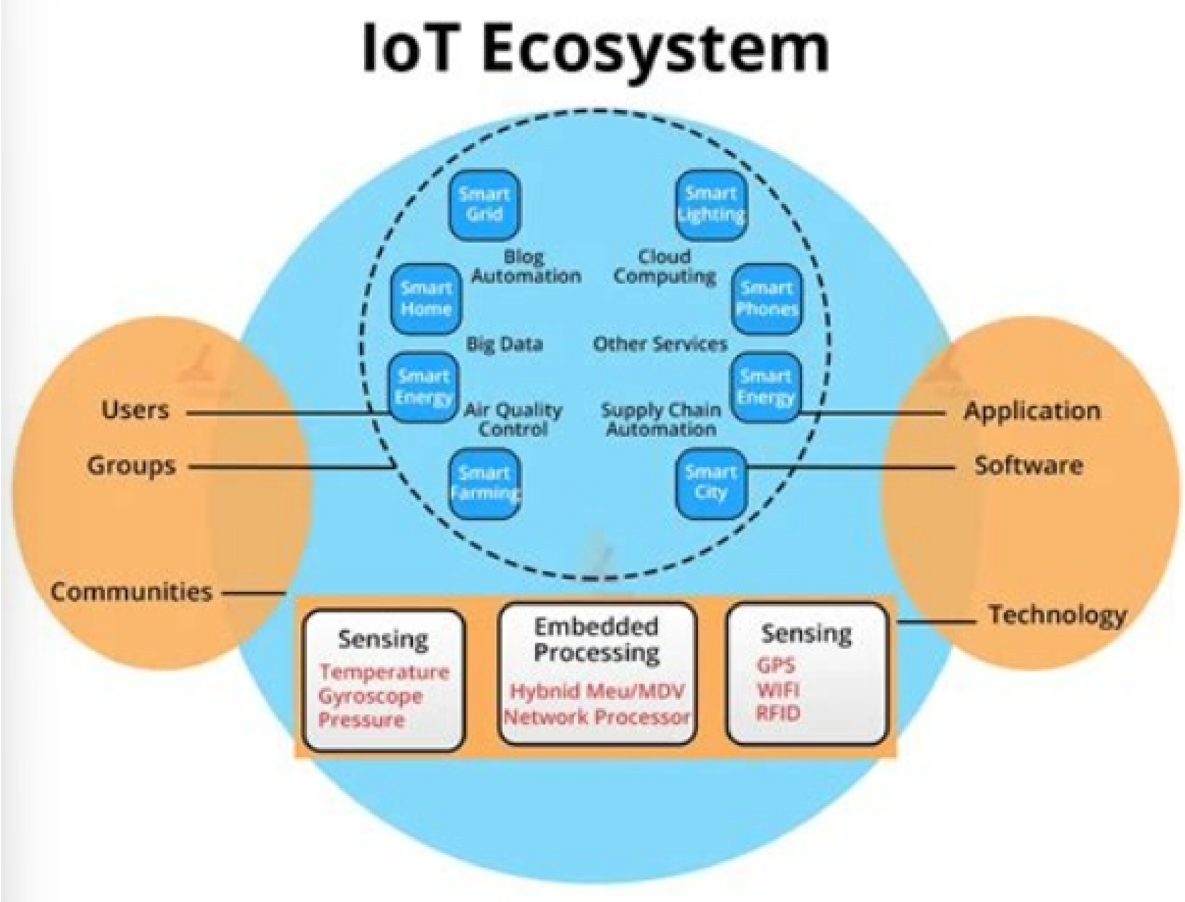


IoT Ecosystem and Architecture

1) Perception/Sensing Layer,

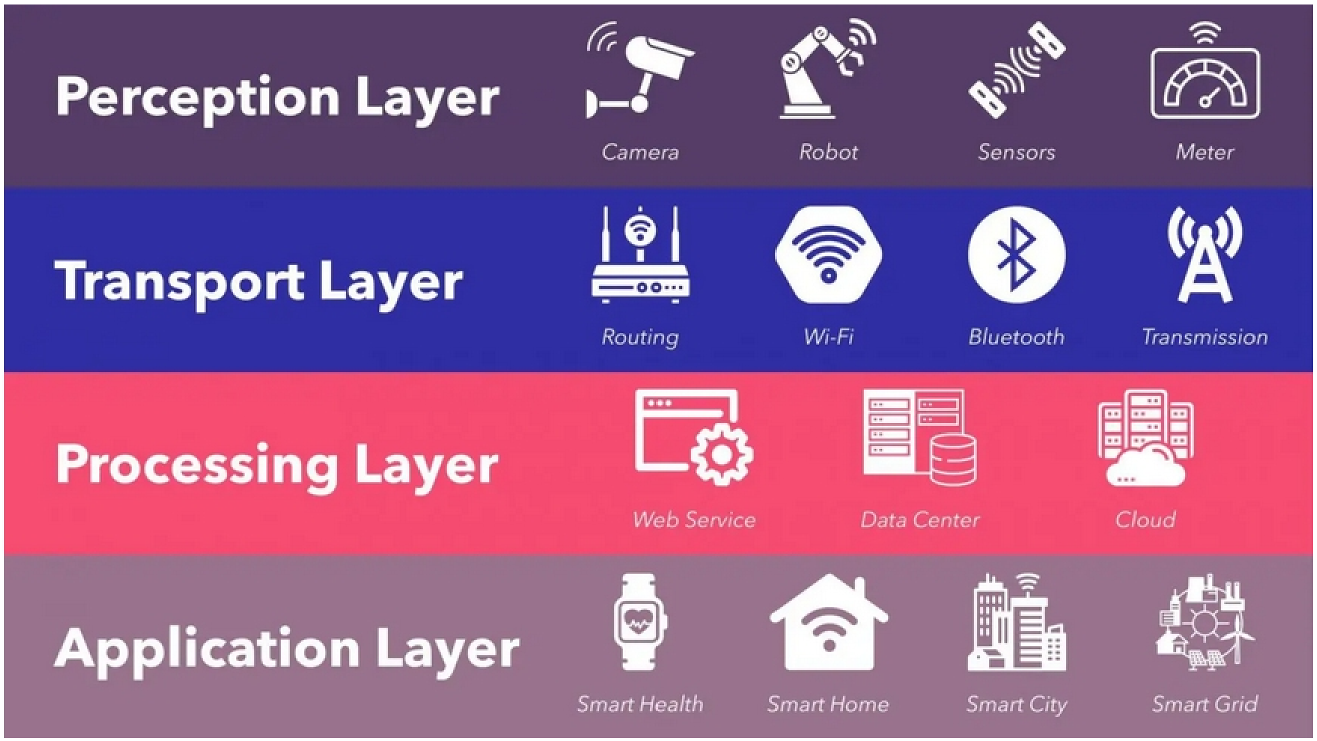
2) Network/Transport Layer,

3) Edge Processing Layer,

4) Cloud Processing Layer, and

5) Application Layer

IoT Ecosystem and Architecture



Sensors and Actuators

- A sensor is a device that detects and measures a physical property from its environment and converts it into a signal, typically an electrical one, that can be read by an observer or an instrument.
- Types of Sensors:
  - Environmental
    - Temperature
    - humidity
    - air quality and
    - gas

Sensors and Actuators

- Types of Sensors:
  - Biometric & Wearable
    - Heart rate,
    - SpO2,
    - electrodermal activity sensors.
  - Positioning
    - GPS,
    - indoor positioning beacons.

Sensors and Actuators

- Actuators are devices that convert electrical signals from IoT systems into physical action.
- They are the output mechanism that allows IoT systems to interact with and control the physical world based on sensor data and intelligent decisions.

Sensors and Actuators

- Types of Actuators:
  - Electrical motors:
    - DC
      - Smart home
      - Personal Mobility
    - Servo
      - Robotics
      - Pet feeder
    - Stepper
      - 3D Printing
      - Robotics

Sensors and Actuators

- Types of Actuators:
  - Solenoid
    - Security system
    - Automotive
    - Vending Machine
  - Hydraulic Actuators
    - Construction Equipment
    - Manufacturing
    - Marine

Sensors and Actuators

- Types of Actuators:
  - Electroactive Polymer
    - Biomedical
    - Microfluidics
  - Voice Coil Actuators
    - Audio System
    - Medical Imaging

Sensors and Actuators

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IoT Hardware: Arduino

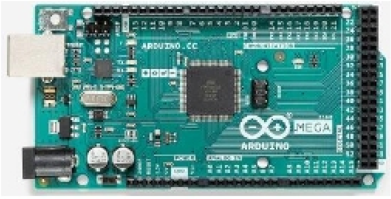
- It is an open-source electronics platform based on easy-to-use hardware and software
- With Arduino, you can control almost everything around you be it simple LED or giant Robots.

DIFFERENT TYPES OF ARDUINO

- **Entry Level** - easy to use and ready to power your first creative projects.
  - Arduino UNO
  - Arduino Nano
  - Arduino Micro

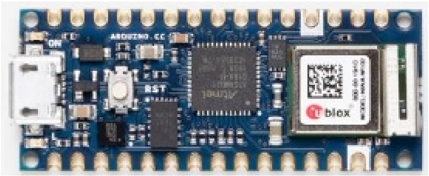
Arduino UNO

- **Enhanced Features** - boards with advanced functionalities, or faster performances
  - Arduino Zero
  - Arduino Mega 2560
  - Arduino Motor Shield



Arduino Mega 2560

- **Internet of Things** - Make connected devices easily with one of these IoT products
  - Arduino Nano 33 IoT
  - Arduino Nano 22 BLE
  - UNO WiFi REV2

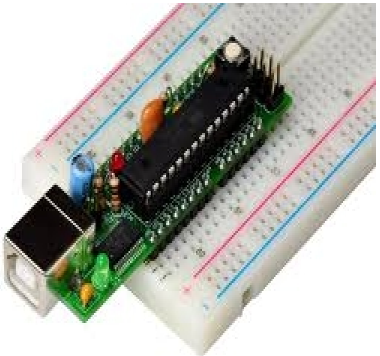


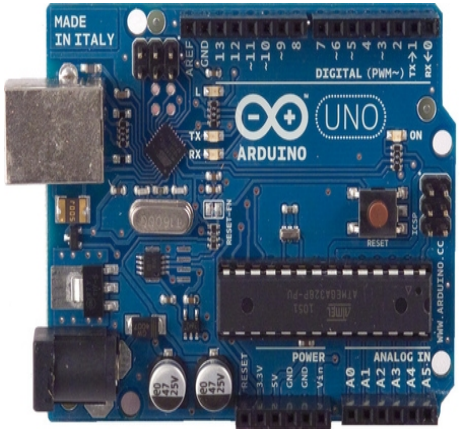
Arduino Nano 33 IoT

Source: https://www.arduino.cc/

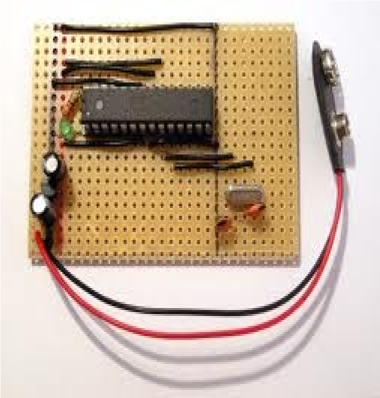
DIFFERENT TYPES OF ARDUINO

Boarduino Kit



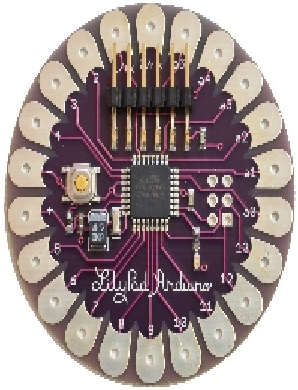


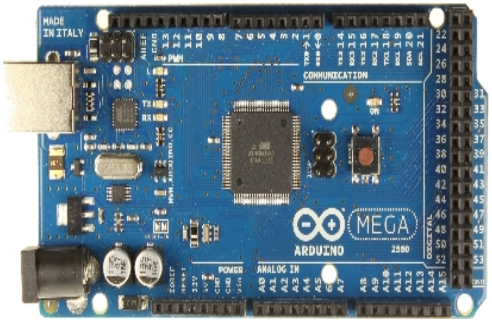
Arduino Uno



DIY Arduino

Arduino LilyPad

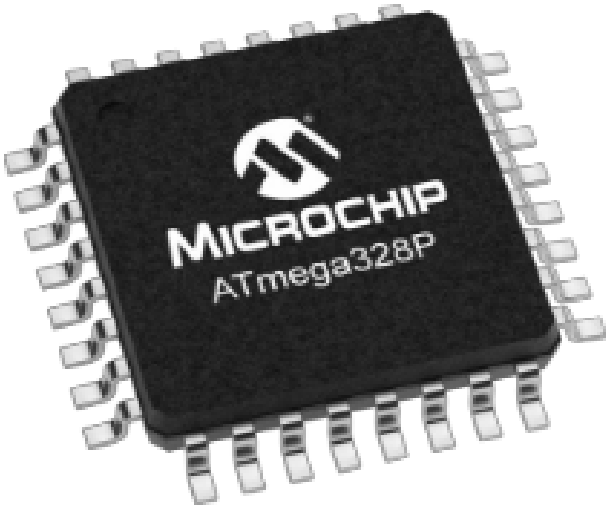




Arduino Mega 2560

ARDUINO UNO

- **Arduino UNO** is a Single board Microcontroller based on **ATmega328P** Processor
  - a product of Atmel (now Microchip)
  - **32** - represents it's flash memory capacity that is 32KB
  - **8** - represents it's CPU type that is of 8 bit
  - **p** - simply denotes picoPower (i.e. very low power).

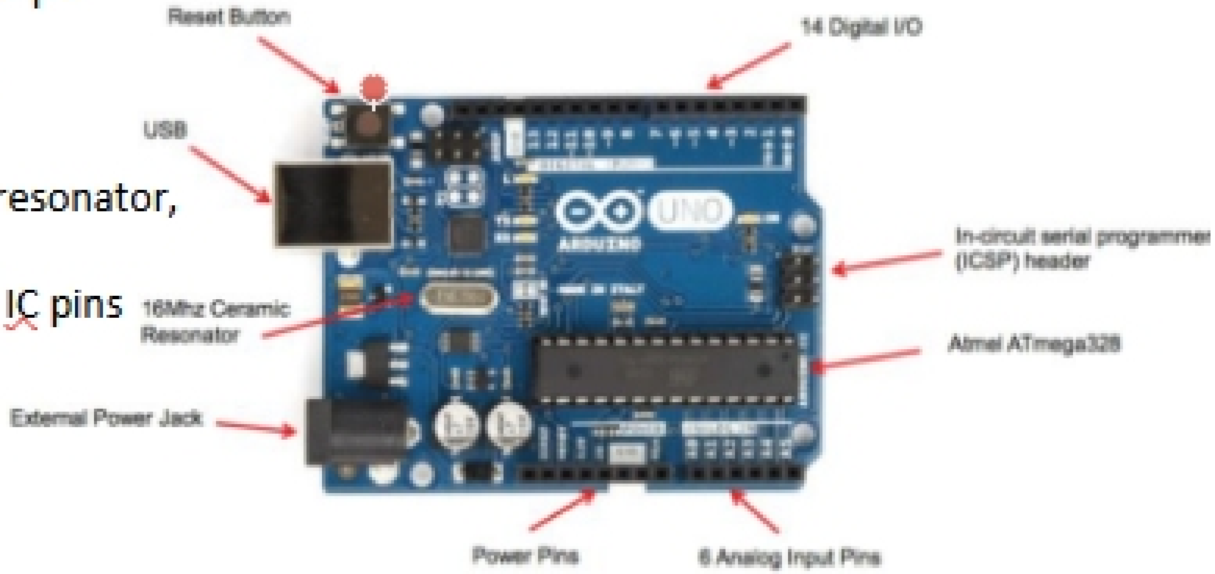


AVR CPU at up to 16 MHz,
32KB Flash, 2KB SRAM, 1KB EEPROM

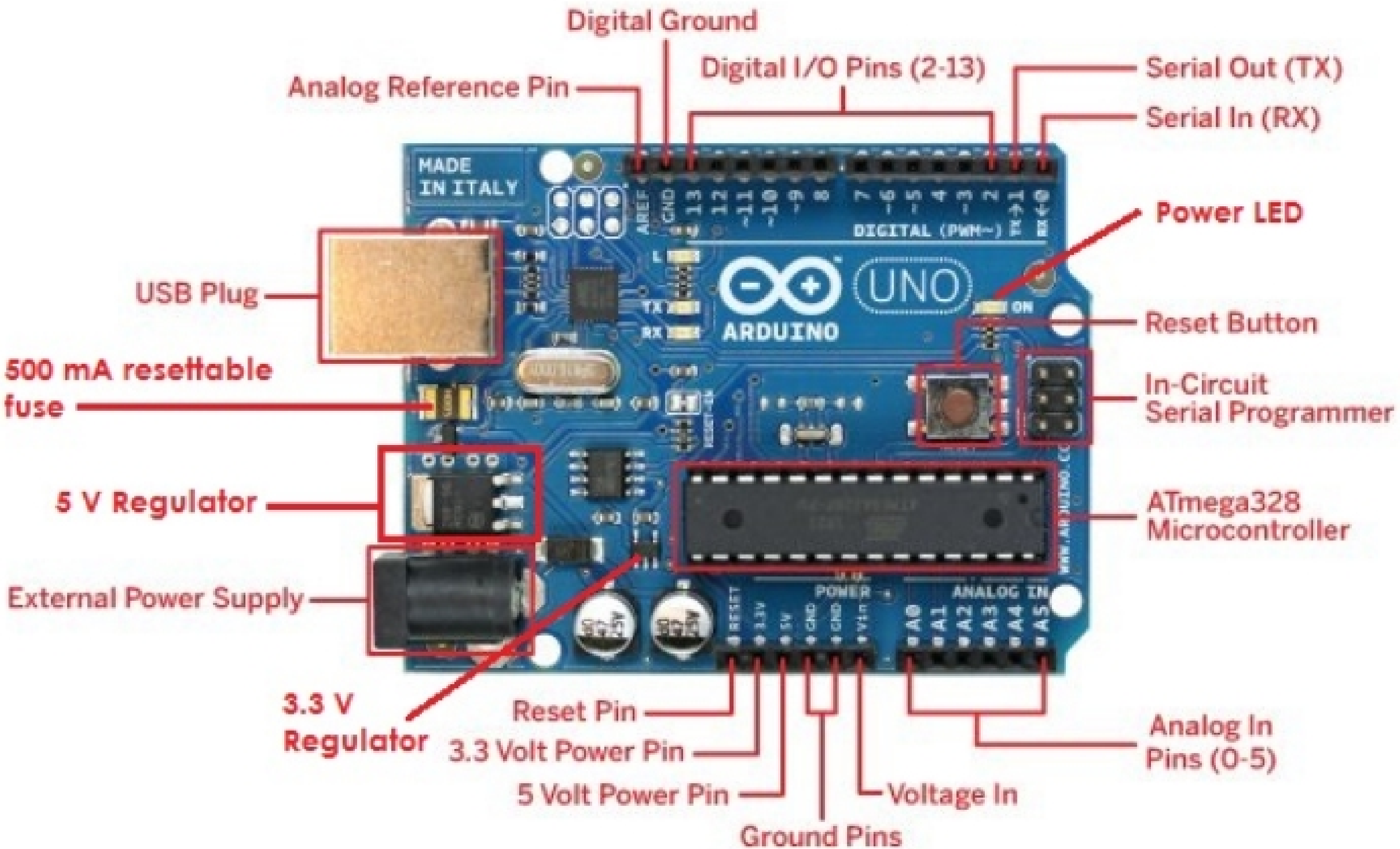
Few competitors: STM32 ARM Cortex , MSP430, and PIC MCU

PINS IN ARDUINO UNO R3

- It has the following major pins/jacks:
  - ✓ 14 digital input/output pins (of which 6 can be used as PWM outputs),
  - ✓ 6 analog inputs,
  - ✓ 6 pins related to energy/power
  - ✓ a reset pin
  - ✓ an analog reference pin
  - ✓ a reset button
  - ✓ a USB connection,
  - ✓ a power jack,
  - ✓ a 16 MHz ceramic resonator,
  - ✓ two ICSP header
  - ✓ Atmel ATmega328 IC pins



DETAILED PIN DIAGRAM



PIN DESCRIPTION

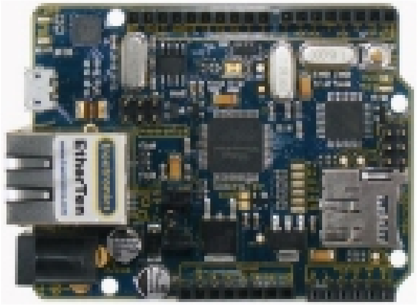
Pin category	**Pin Name**	**Details**
Power Pins	Vin, 3.3V, 5V, GND, RESET	**Vin** : Input voltage to Arduino when using an external power source. **5V** : Regulated power supply used to power microcontroller and other components on the board. **3.3V** : 3.3V supply generated by on-board voltage regulator. Maximum current draw is 50mA. **GND** : ground pins. **Reset**: Reset the microcontroller
ICSP: In-Circuit Serial Programming	ICSP pins: MISO, VCC, SCK, MOSI, RESET, GND	Used to code and boot an Arduino from an external source. Allow inter workings of two or more Arduino boards. Allow you to upload your firmware.

PIN DESCRIPTION

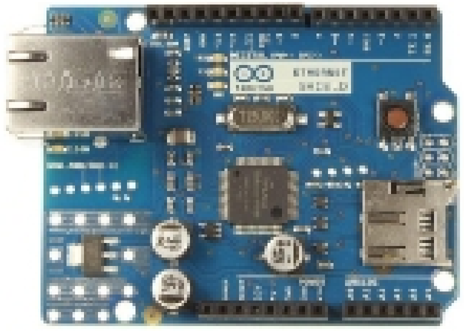
Pin category	Pin No / name	Details
Analog pin	A0 - A5	Used to provide analog input in the range 0-5V.
Digital Input/output pin	Digital Pins 2 - 13	Can be used as input or output pins.
Serial Communication	0(Rx),1(Tx)	Used to receive and transmit TTL serial data.
External Interrupts	2, 3	To trigger an interrupt.
PWM: Pulse Width Modulation	3, 5, 6, 9, 10, 11	Provides 8-bit PWM output.
SPI: Serial Peripheral Interface	10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK)	Used for SPI communication.
Inbuilt LED	13	To turn on the inbuilt LED.
I2C: Inter-IC, or TWI: Two Wire Interface	A4 (SDA: Serial Data), A5 (SCL: Serial Clock)	Used for TWI / I2C communication.
AREF	AREF : Analog Reference Voltage	To provide reference voltage from an external power supply for analog-to-digital conversion of inputs to the analog pins. E.g. if AREF is 4V – the analogRead() range of 0~1023 will relate to 0~4V and not 0~5V.

ARDUINO IN IOT

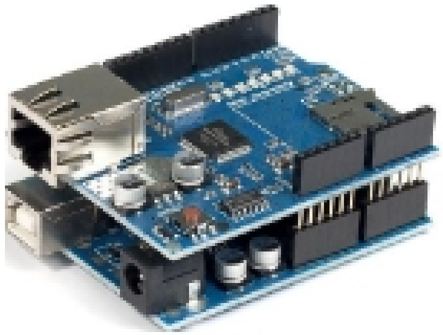
- Arduinos are used to create IoT projects.
- But, it requires either a specialized Arduino or shields to provide network capabilities
- The **network interface** could be Ethernet / WiFi / Cellular



EtherTen



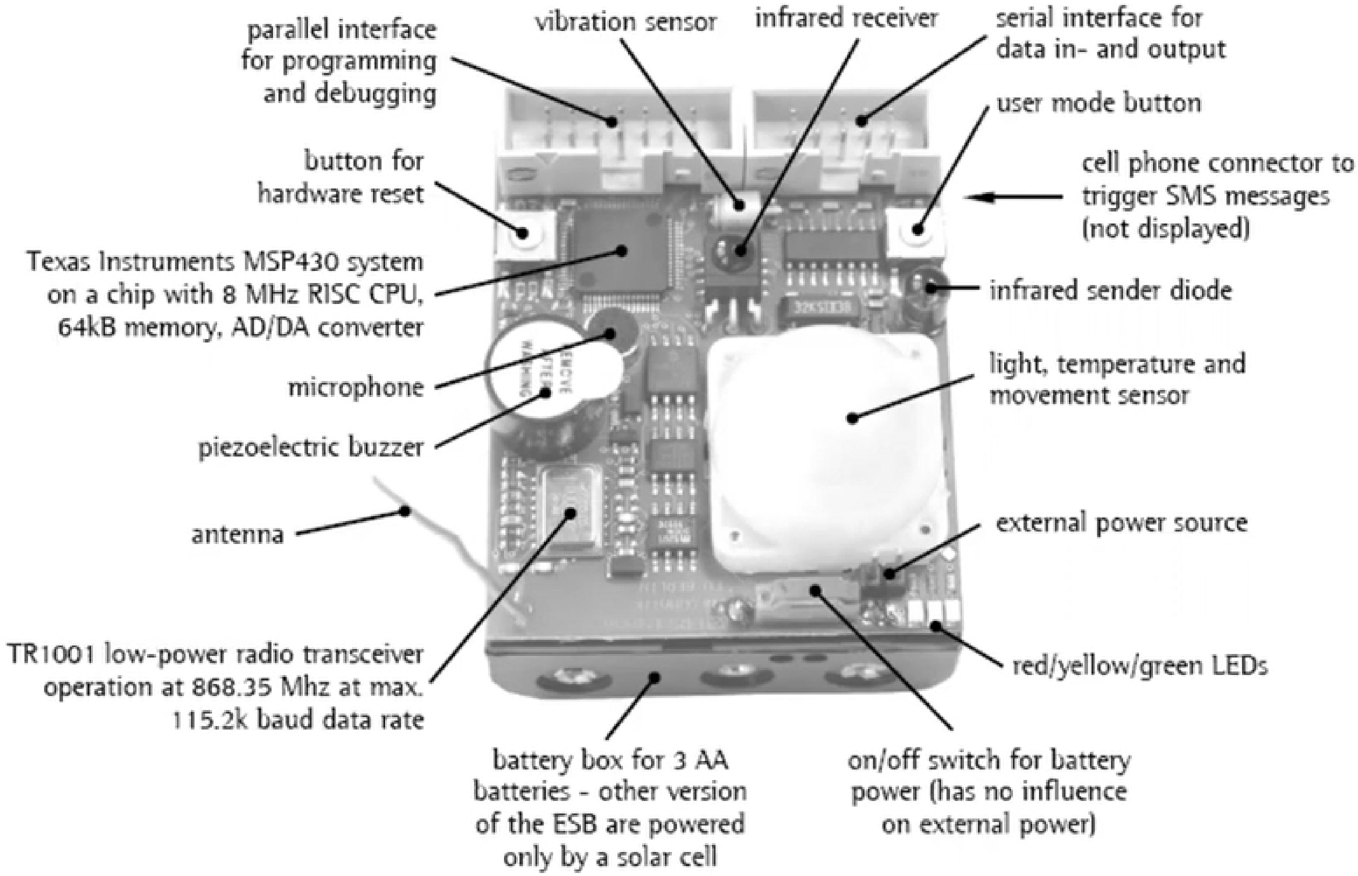
Arduino Ethernet Shield



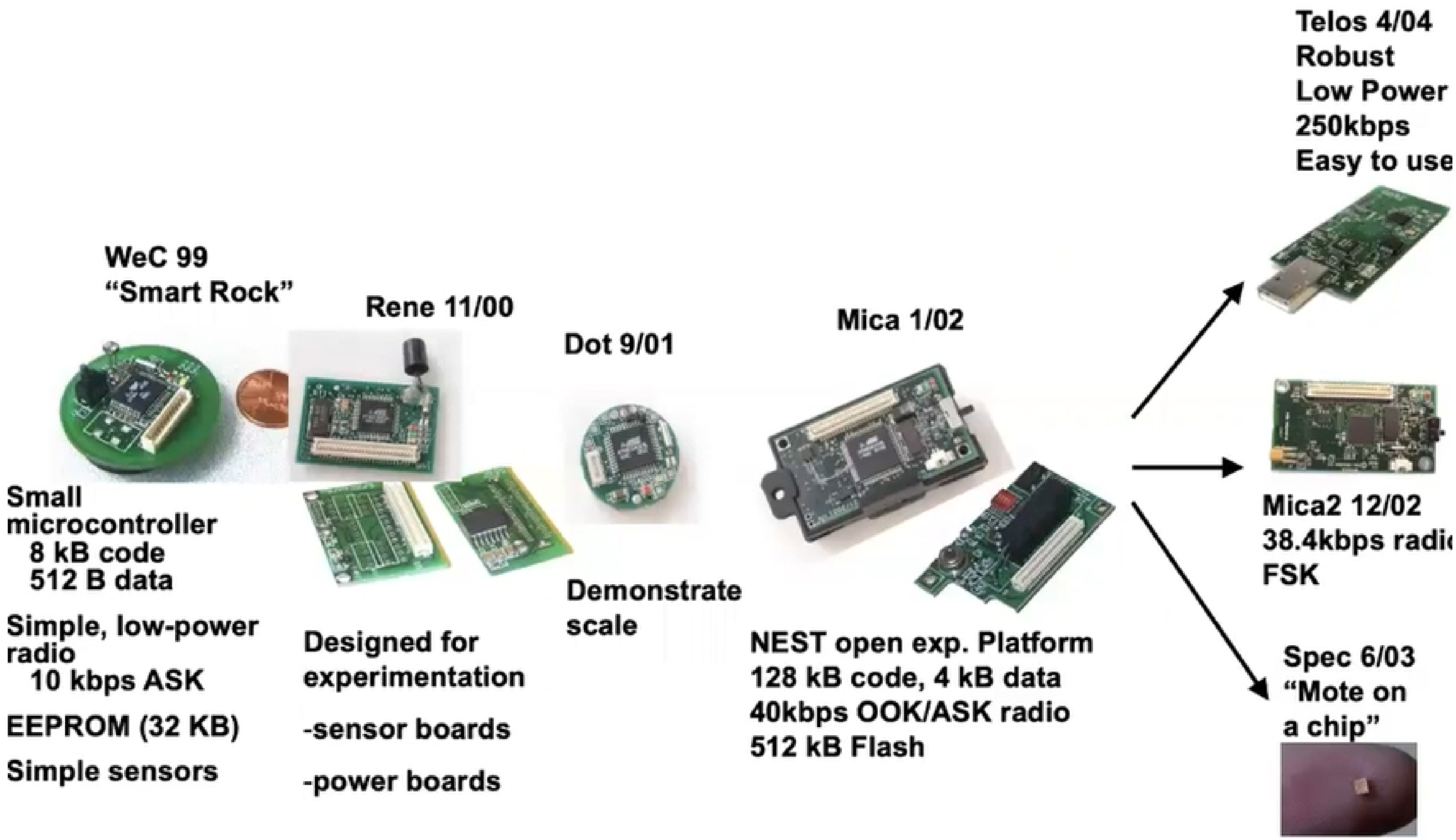
Arduino + Ethernet Shield

Arduino UNO WiFi Rev2

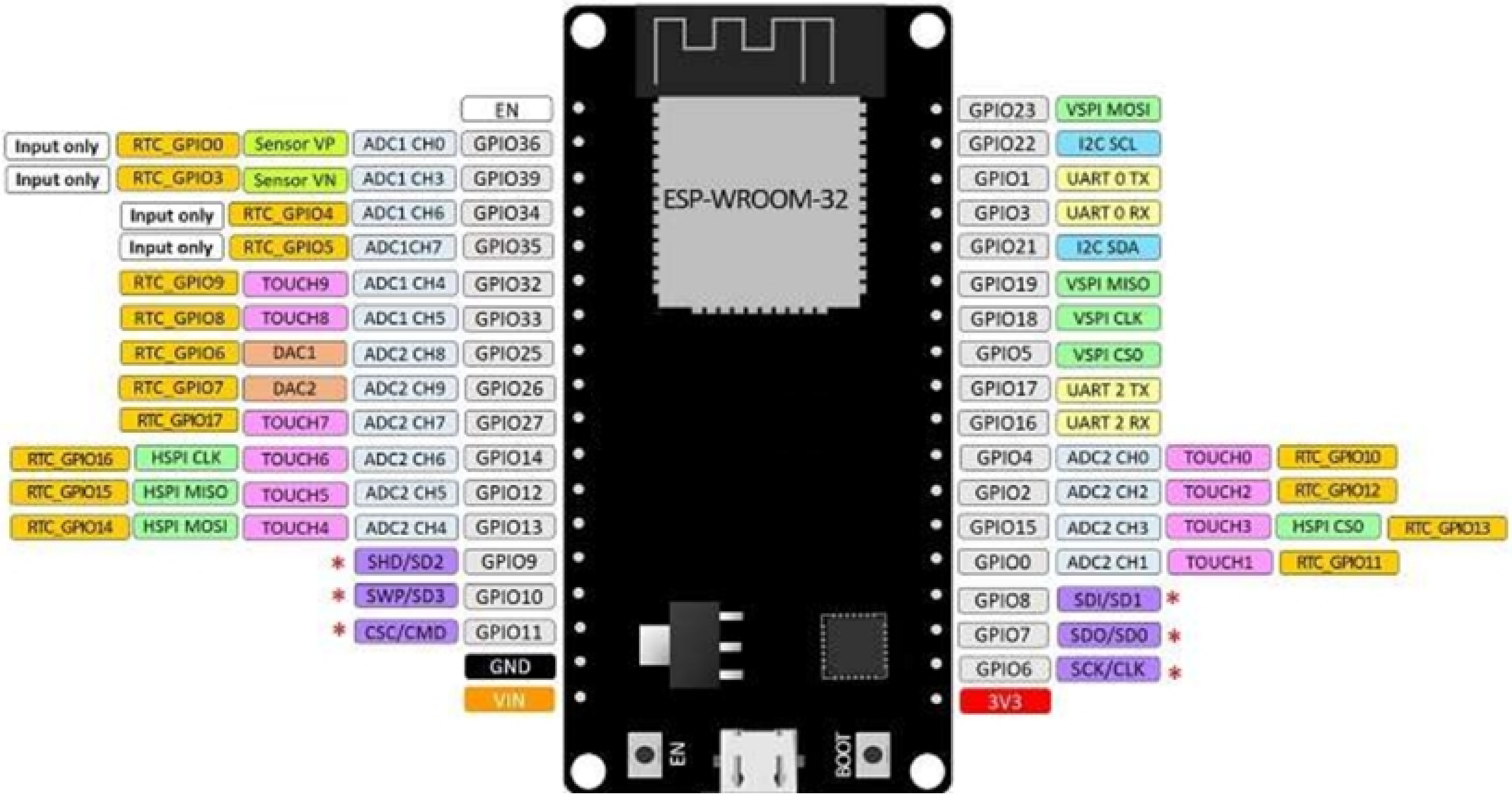
ARDUINO IN IOT



ARDUINO IN IOT



ESP32 Pins


ESP-WROOM-32
EN
Input only RTC_GPIO0 Sensor VP ADC1 CH0 GPIO36
Input only RTC_GPIO3 Sensor VN ADC1 CH3 GPIO39
Input only RTC_GPIO4 ADC1 CH6 GPIO34
Input only RTC_GPIO5 ADC1CH7 GPIO35
RTC_GPIO9 TOUCH9 ADC1 CH4 GPIO32
RTC_GPIO8 TOUCH8 ADC1 CH5 GPIO33
RTC_GPIO6 DAC1 ADC2 CH8 GPIO25
RTC_GPIO7 DAC2 ADC2 CH9 GPIO26
RTC_GPIO17 TOUCH7 ADC2 CH7 GPIO27
RTC_GPIO16 HSPI CLK TOUCH6 ADC2 CH6 GPIO14
RTC_GPIO15 HSPI MISO TOUCH5 ADC2 CH5 GPIO12
RTC_GPIO14 HSPI MOSI TOUCH4 ADC2 CH4 GPIO13
* SHD/SD2 GPIO9
* SWP/SD3 GPIO10
* CSC/CMD GPIO11
GND
VIN
GPIO23 VSPI MOSI
GPIO22 I2C SCL
GPIO1 UART 0 TX
GPIO3 UART 0 RX
GPIO21 I2C SDA
GPIO19 VSPI MISO
GPIO18 VSPI CLK
GPIO5 VSPI CS0
GPIO17 UART 2 TX
GPIO16 UART 2 RX
GPIO4 ADC2 CH0 TOUCH0 RTC_GPIO10
GPIO2 ADC2 CH2 TOUCH2 RTC_GPIO12
GPIO15 ADC2 CH3 TOUCH3 HSPI CS0 RTC_GPIO13
GPIO0 ADC2 CH1 TOUCH1 RTC_GPIO11
GPIO8 SDI/SD1 *
GPIO7 SDO/SD0 *
GPIO6 SCK/CLK *
3V3
EN
BOOT

ESP32 Pins

GPIO	Input	Output	Notes
0	pulled up	OK	outputs PWM signal at boot
1	TX pin	OK	debug output at boot
2	OK	OK	connected to on-board LED
3	OK	RX pin	HIGH at boot
4	OK	OK	
5	OK	OK	outputs PWM signal at boot
6	x	x	connected to the integrated SPI flash
7	x	x	connected to the integrated SPI flash
8	x	x	connected to the integrated SPI flash
9	x	x	connected to the integrated SPI flash
10	x	x	connected to the integrated SPI flash
11	x	x	connected to the integrated SPI flash
12	OK	OK	boot fail if pulled high
13	OK	OK	
14	OK	OK	outputs PWM signal at boot
15	OK	OK	outputs PWM signal at boot
16	OK	OK	

ESP32 Pins

17	OK	OK	
18	OK	OK	
19	OK	OK	
21	OK	OK	
22	OK	OK	
23	OK	OK	
25	OK	OK	
26	OK	OK	
27	OK	OK	
32	OK	OK	
33	OK	OK	
34	OK		input only
35	OK		input only
36	OK		input only
39	OK		input only

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Thanks